Institute of Technology of Cambodia

Department of Information and Communication Engineering

Assignment: TP10

Subject: **Operating System**

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Practice Exercise

Case Scenario 1

There has been suspicious activity on the system. In order to preserve log information, it will be necessary to archive the current files in /var/log ending with the .log extension. The files are to be saved to a file named log.tar, stored in the directory, ~/archive. It has also been requested that the files that were archived be saved to a directory, ~/backup.

Objectives

• Create an archive named `log.tar` that is stored in the archive directory located in thr home directory.
• Remove path names from the files that are archived.
• Produce verbose output while archiving.
• List the contents of the archive without extracting.
• Extract the files to the directory, `~/backup`

```
sokmeak@sokmeaksaren:~/archive$ tar -cvf ~/archive/log.tar --transform='s:.*/::' /var/log/*.log
tar: Removing leading `/' from member names
/var/log/alternatives.log
tar: Removing leading `/' from hard link targets
/var/log/apport.log
/var/log/auth.log
/var/log/bootstrap.log
/var/log/cloud-init.log
/var/log/cloud-init-output.log
/var/log/dpkg.log
/var/log/fontconfig.log
/var/log/kern.log
/var/log/ubuntu-advantage-apt-hook.log
```

```
sokmeak@sokmeaksaren:~/archive$ tar -tvf ~/archive/log.tar
-rw-r--r-- root/root         0 2025-01-01 02:16 alternatives.log
-rw-r----- root/adm          0 2024-12-27 07:05 apport.log
-rw-r----- syslog/adm    50895 2025-01-17 10:18 auth.log
-rw-r--r-- root/root     61237 2024-08-27 14:19 bootstrap.log
-rw-r----- syslog/adm    84520 2024-12-27 07:05 cloud-init.log
-rw-r----- root/adm       4621 2024-12-27 07:05 cloud-init-output.log
-rw-r--r-- root/root         0 2025-01-01 02:16 dpkg.log
-rw-r--r-- root/root      4659 2024-12-27 07:22 fontconfig.log
-rw-r----- syslog/adm   364568 2025-01-17 10:16 kern.log
-rw-r--r-- root/root         0 2024-12-27 07:10 ubuntu-advantage-apt-hook.log
sokmeak@sokmeaksaren:~/archive$ mkdir -p ~/backup
sokmeak@sokmeaksaren:~/archive$ tar -xvf ~/archive/log.tar -C ~/backup
alternatives.log
apport.log
auth.log
bootstrap.log
cloud-init.log
cloud-init-output.log
dpkg.log
fontconfig.log
kern.log
ubuntu-advantage-apt-hook.log
sokmeak@sokmeaksaren:~/archive$ cd ..
sokmeak@sokmeaksaren:~$ ls backup/
alternatives.log  auth.log       cloud-init.log         dpkg.log        kern.log
apport.log        bootstrap.log  cloud-init-output.log  fontconfig.log  ubuntu-advantage-apt-hook.log
sokmeak@sokmeaksaren:~$
```

Case Scenario 2

1. Create a simple scheduled job that executes the date command every Friday at 10:31 am. I can redirect the result of this job to another terminal with location `/dev/pts/1`

```
31 10 * * 5 date >/dev/pts/1
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
```

```
sokmeak@sokmeaksaren: ~        sokmeak@sokmeaksaren: ~
Hello sokmeak! How are you today?
sokmeak@sokmeaksaren:~$ tty
/dev/pts/1
sokmeak@sokmeaksaren:~$ Hello
Hello
Fri Jan 17 10:31:01 AM UTC 2025
```

2. Create another scheduled job that executes a simple script with date command every minute, redirecting the output to the output.log file in your home directory. >> crontab -e

```
* * * * * ~/time_trigger
```

```
sokmeak@sokmeaksaren: ~                                    sokmeak@sokmeaksaren: ~
  GNU nano 7.2                                                time_trigger
#!/bin/bash

date >> ~/output.log
```

```
sokmeak@sokmeaksaren:~$ cat output.log
Fri Jan 17 10:40:01 AM UTC 2025
Fri Jan 17 10:41:01 AM UTC 2025
Fri Jan 17 10:42:01 AM UTC 2025
Fri Jan 17 10:43:01 AM UTC 2025
```

3. Create a schedule job that archive a folder backup in your home directory every week on Monday 12:00 am.

```
sokmeak@sokmeaksaren: ~                                    sokmeak@sokmeaksaren: ~
  GNU nano 7.2                                        /tmp/crontab.MsQh0R/crontab
18,21,22,23 7 * * 5 date >/dev/pts/1
* * * * * ~/time_trigger

0 0 * * 1 tar -zcvf ~/backup_$(date +\%Y\%m\%d).tar.gz ~/backup
```